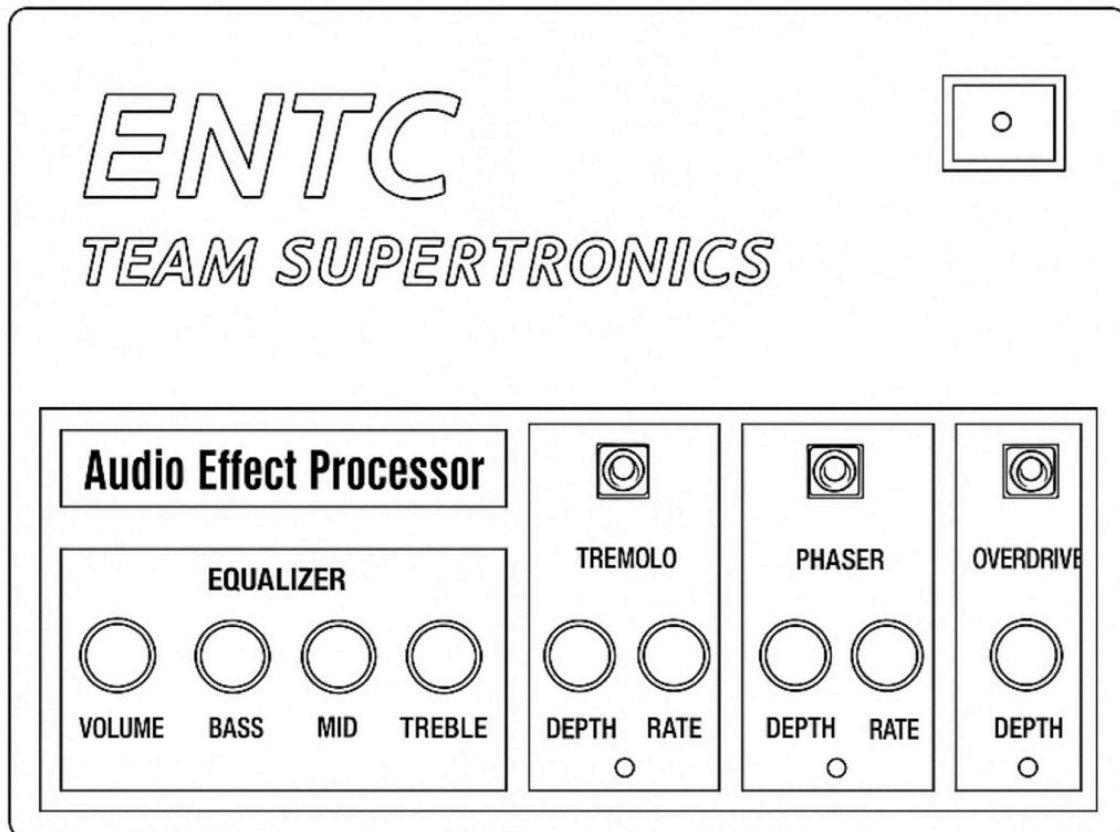


Analog Audio Effect Processor



User Manual

Designed and developed by
Team Supertronics,
University of Moratuwa, Sri Lanka

1 Introduction

The Analog Audio Effect Processor is a fully analog signal-processing device designed for real-time audio manipulation. The system integrates multiple classic audio effects using operational amplifiers, passive components, and discrete analog circuitry. It supports both mono and stereo inputs while always producing a mono output.

This device is suitable for guitars, musical instruments, laptops, mobile phones, and general audio sources.

2 Safety Information

- Operates from **230–240 V AC, 50 Hz**. Do not open the enclosure while powered.
- Not battery powered.
- Reverse-polarity protection is internally provided.
- No user-serviceable parts inside.
- Use only standard audio cables and approved adapters.

3 Power Supply

The device contains an internal regulated dual-supply analog power system. Power is switched using the rear-panel rocker switch. Status LEDs indicate active modules only, not power faults.

4 Input and Output Connections

4.1 Audio Input

- Connector: 3.5 mm audio jack
- Mono or stereo input accepted
- Stereo inputs are internally summed to mono

4.2 Audio Output

- Connector: 3.5 mm audio jack
- Mono output signal
- Compatible with earphones, powered speakers, amplifiers, and audio interfaces

5 Front Panel Controls

5.1 Equalizer Section

- **Volume:** Master output level
- **Bass:** Low-frequency control
- **Mid:** Mid-frequency shaping
- **Treble:** High-frequency emphasis

5.2 Overdrive

- Toggle switch enables/disables distortion
- Depth knob controls clipping intensity
- LED indicates active overdrive

5.3 Tremolo

- Depth control sets modulation depth
- Rate control sets modulation speed
- LED indicates tremolo activity

5.4 Phaser

- Depth controls phase shift intensity
- Rate controls sweep frequency
- LED indicates active phaser

6 How to Use the Device

1. Connect the audio source to the input jack.
2. Connect earphones, speakers, or amplifier to the output.
3. Power on the device using the rear switch.
4. Enable desired effects using individual toggle switches.
5. Adjust knobs gradually to shape the sound.
6. Combine multiple effects as required.

7 Supported Audio Sources

- Electric and acoustic guitars
- Musical instruments with line output
- Laptop and mobile phone audio
- Recorded audio and general line-level signals

8 Handling and Maintenance

- Operate using hand controls only
- Avoid excessive heat and moisture
- Clean with a dry cloth
- Do not apply mechanical force to knobs or switches

9 Disclaimer

This device is an academic and experimental analog electronics project. Performance may vary depending on source impedance, load conditions, and external equipment.

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